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Chapter 5 Maintenance Instruction of Hydraulic System

Summary：

This QC maintenance instruction covers three parts: mechanical, electric and hydraulic. Hydraulic systems are applied diffusely in our Quayside Crane (QC), they have a series of advantages, such as: small volume, light quantity, compact configuration, stead working, self-lubricated and long life of working parts, etc. In the QC, there are different kinds of hydraulic system used in disc brake, wheel brake and multi-function system.

To make sure the exertion of hydraulic system and high level completion, it is important to deal with the maintenance and fault diagnosis. The maintenance is based on equipment state monitoring, then make prevention service according to state information to reduce or avoid fault, keep the hydraulic system in a normal good state .As a result ， the safety and high efficiency of QC can be guaranteed.

As followed contents , just for reference, if you need the detail operation and maintenance instruction, please review the Hydraulic maintenance manual.

5.1 General

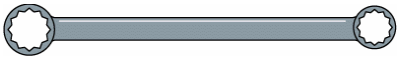
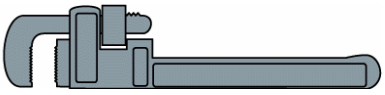
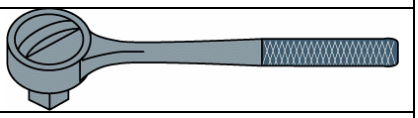
5.1.1 Attention Note

The front part of this chapter is about the function of different hydraulic system, and follow part, is about the maintenance and fault diagnosis. Especially to point out that there should be professional person to be in charge of the safety.

	**Only professional person can do the service**
	**Study this booklet before service**
	**Avoid hand clamping**

5.1.2 Common tools of hydraulic maintenance

Table 5-1：Tool list

Inner hexagon spanner		Double offset ring spanner	
Amphibious spanner		Screw driver	
Tube spanner		Female spanner	
Pressure gauge		Vernier caliper	
Tacho meter		Thermo meter	
multimeter		Common electrician tool	

5.2 Common parts of hydraulic system

5.2.1 Hydraulic pump

All hydraulics pump of the crane in the item are variable displacement axial piston pump (see picture as followed)



Fig.5-1：Variable displacement axial piston pump

Structure drawing **Fig.5-2：**

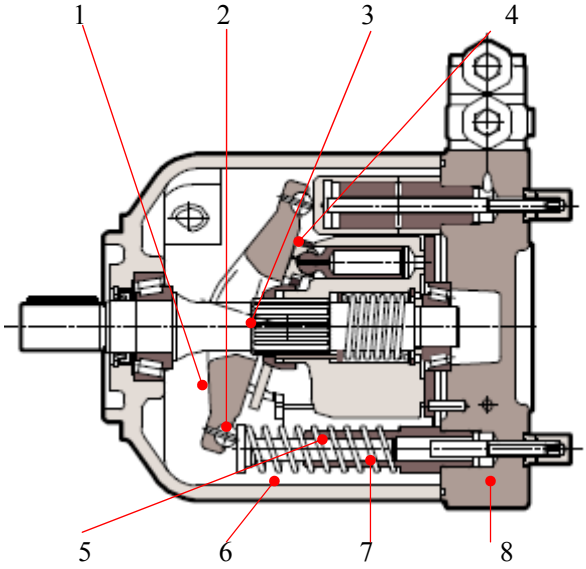


Fig.5-2：Inner structure schematic drawing

1 swash plate 2 Skid device 3 drive shaft 4 backhaul plate
5 piston 6 barrel 7 spring 8 valve plate

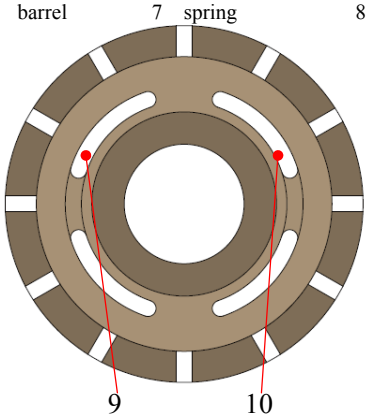


Fig.5-3： valve plate

9 drain window 10 suction window

The piston pump consists of drive shaft 3,barrel 6,valve plate8, piston 5 and swash plate 1.drive shaft 3 drive the barrel 6 to rotate through gear on the shaft, there are 7 piston holes on the barrel equally, and with piston 5 axial installed in. when barrel rotate, the pistons will rotate with the barrel and slide in the piston hole as well at the same time.(under the pressure and spring force, pistons are against swash plate1 tightly) There are a drain window 9 and a suction window 10 that are camber on valve plate 8,the swash plate 1 is mounted with a angle r with the axial of barrel 6. Valve plate 8 and swash plate 1 are fixed, when barrel 6 rotate, under the effect of swash plate, push the piston do the alternative motion, then make the oil suction and drain procession through plate window 9,10.

When barrel 6 motions from top to bottom of the housing shown in Fig. 5-2, piston 5 (under the spring force : the spring force pass to piston through backhaul plate 4 and valve insert 2)will extend, then the sealed volume of the piston hole will expand and form a partial vacuum. The oil in tank will be sucked into piston hole through the suction window 10 on valve plate 8, which means the suction process. When barrel 6 do the motion from bottom to top of the housing shown in Fig. 5-2, piston 5 (under the spring force : the spring force pass to piston through backhaul plate 4 and valve insert 2)will retract into piston hole, then the sealed volume of the piston hole will reduce and oil will be pushed out. The oil in tank will be drained out to tube system through the drain window 9 on valve plate 8, which means the drain process. Every piston will finish suction and drain procession when barrel 6 turns a circle. So if the barrel continuously rotates, pistons will continuously suck and drain oil. This is the working principle of axial piston pump.



Fig.5-4：Piston pump 3D drawing

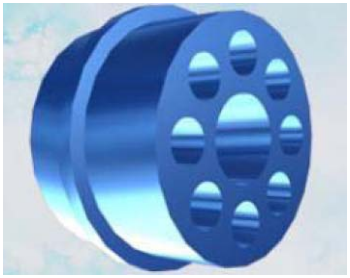


Fig.5-5：Barrel 3D drawing



Fig.5-6：Valve plate drawing

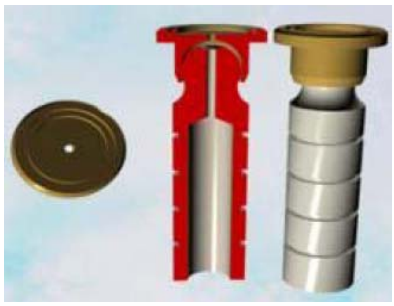


Fig.5-7：Piston 3D drawing

As Fig. 5-8 shows following are the expression of piston pump in hydraulic system. The pressure is controlled by fixed value, and it can be adjusted by pilot valve. The pump will stop when pressure is up to the preset value.

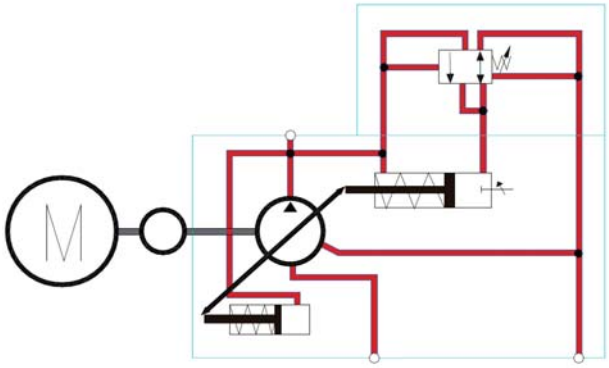


Fig.5-8：Piston pump principle drawing

Table 5-2：Common Fault and Measurement of Pump

Fault Phenomenon	Reason	Measurement
Not enough flux or no oil drain	Flux adjuster fault or swash plate angle too small	Fix the adjuster or adjust the angle
	Backhaul plate broken then pump can't suck	Replace the plate
	Spring broken then can't go back or not enough, loose seal between barrel and valve plate	Replace the spring
insufficient output pressure	Air in pilot valve or valve blocked	Vent air or clean the pilot valve
	Outer leakage	Fasten the joint or replace the seals.
Flux adjuster fault	Jam in orifice of control line	Filter the oil and clean the orifice
	Piston or spring blocked	Grind measurement if block; Clean the oil when pollution
Difficult to rotate	Piston or barrel blocked; ball top of piston broken or insert broken	Clean and replace the parts and reassemble
Heavy noise	Air in tube, oil polluted, high viscosity or leakage、 not concentric with the motor's shaft Pipe vibration	Vent air, change and clean oil, fasten the tube connector Adjust the assemble parts Reduce the vibration
System too hot	Highly inner leakage or small volume	Check the seal surface then fix it

5.2.2 Hydraulic cylinder

There are two kinds of cylinder in QC, one is single-act cylinder which is reverted by spring, such as wheel brake and disc brake; the other kind is single rod cylinder. Cylinder like snag cylinder is dual-act single rod cylinder and steel rope tension cylinder is single-act one. The maintenance of the first kind can be found in brake manual. The section here is just about constitution principles and maintenance instruction of the second kind



Fig.5-9： T/L/S and snag cylinder



Fig.5-10：Tension cylinder

Cylinder working principle: because of the effective forcing area and oil pressure difference, piston will do the alternative motion under the hydraulic power (PASCAL PRINCIPLE)

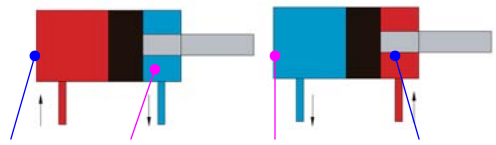


Fig.5-11：Cylinder's working principle

1 non-rod chamber 2 rod chamber

Cylinder structure is shown as Fig.5-12，

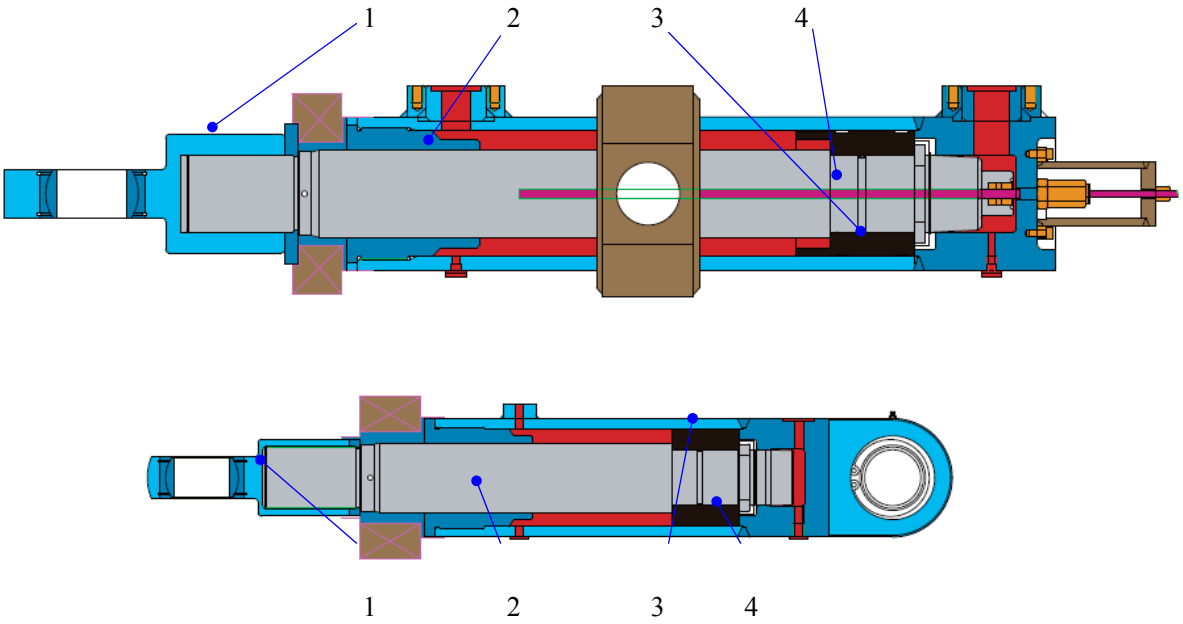


Fig.5-12：Cylinder inner structure

(top one is snag cylinder，bottom one is tension cylinder)

1 cover 2 rod 3 piston chamber 4 barrel

Table 5-3：Common fault and measurement of cylinder

Fault phenomenon	reason	measurement
Stick-lip	Air in barrel	Use air vent valve to vent air
	Too much compress of movable seal	Adjust seal, check the groove size, make sure 5 bar as start pressure
	Piston is not coaxial with rod	Adjust the rod ,make coaxial coefficient <ϕ0.04mm
	Piston rod distortion	Adjust the rod beeline < 0.1/1000
	Heavy abrasion of barrel surface	Fix the barrel or replace
	Assembly fault	Adjust
	Too tight of rod nut	adjust
	Bad lubrication of orbit	Improve the lubrication
weak push force, low speed or speed loosing	Big gap between barrel and piston or o ring broken	Replace the o ring and adjust the gap
	Leakage by rod distortion	Fix or replace the rod or barrel
	Too tight of seal make too heavy friction	Loose, adjust or replace the seal
	Rod distortion make heavy friction	Adjust the rod
impact	Big gap between barrel and rod or throttle valve fault	Replace or fix the rod ,check the throttle valve
	Check valve fault on top buffer device	Fix or replace the check valve or valve base
Outer leakage	Seal broken or bad assembly	Replace the seal and re-assembly
	Rod surface seizure	Check and fix the rod
	Seal broken of joint or connector	Check and fix
	Bad seal of cylinder cover	Check and fix the seal or seal surface

5.2.3 Hydraulic valves

- Valves in QC hydraulic system can be divided into :
- Direction control valve: such as check valve and solenoid valve, etc. used to control the flow direction so that make the start, stop change direction and other function of the executive element.
- Pressure control valves: such as relieve valve, pressure reduce valve and logic valve, etc. are used to control the system pressure so as to satisfying the qualified pressure, torque or working order.
- Flux control valve: such as throttle valve and speed regulator valve, etc. used to control the flux so that make sure the speed qualification of the executive element.

Following are the details:

5.2.3.1 CHECK VALVE



Fig.5-13：Check valve

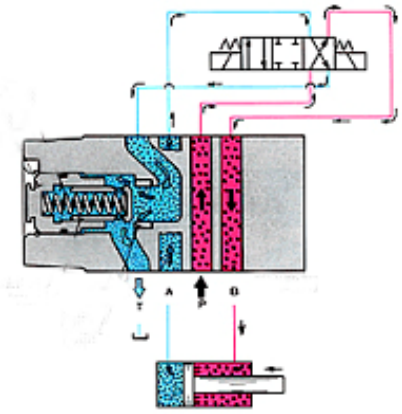


Fig.5-14：Check valve work principle

- As stop backhaul valve, mounted on the outlet of pump. For the safety purpose, check valve is mounted on the outlet of pump to avoid pump from doing reversal rotate then been broken.

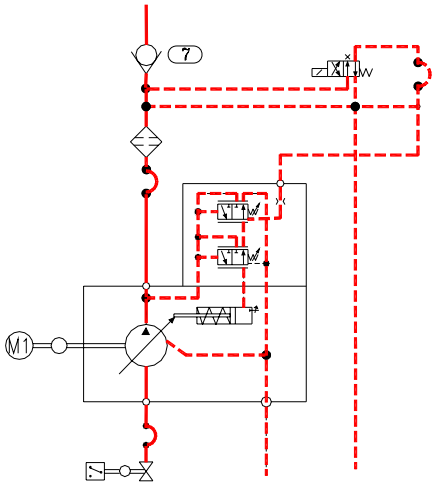


Fig.5-15： real example of check valve

- Combine with count-balance valve to form multi-function valve. In back girder hydraulic system, this set of valves is used to build back-up pressure in drain line, but no limit in suction line. Make sure the cylinder move placidly with changing load, avoid quickly sliding.

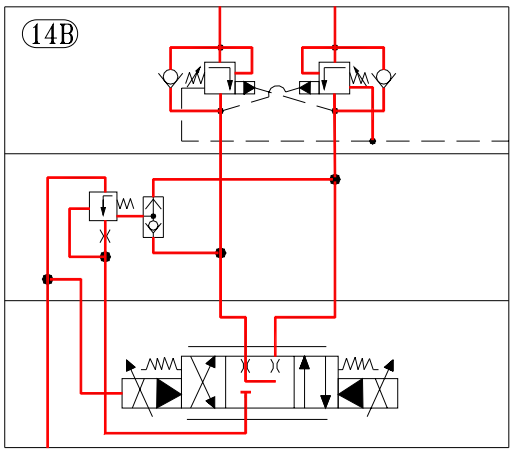


Fig.5-16：combine with count-balance valve

- combine with throttle valve, in the gantry wheel brake system, this set of valves are used to open the wheel brake quickly when high pressure oil has been pushed in, but clamp time are controlled by throttle valve

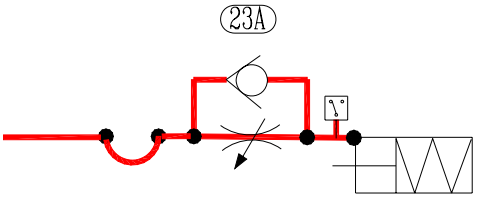


Fig.5-17：Check valve combine with throttle valve

5.2.3.2 SOLENOID VALVE

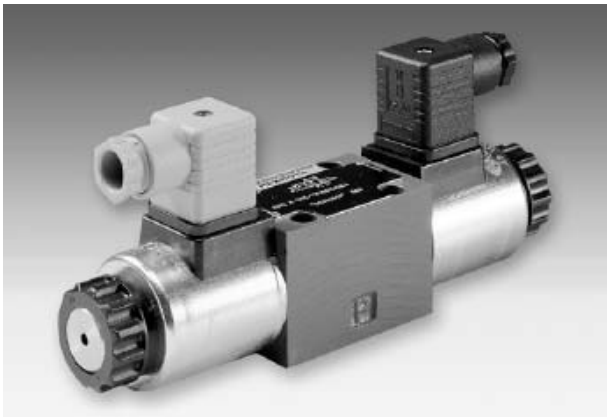


Fig.5-18：Solenoid direction valve

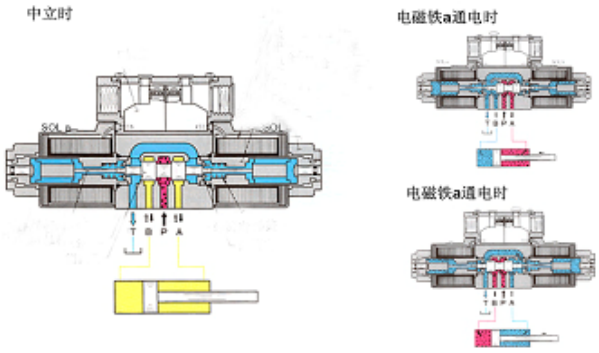


Fig.5-19： Solenoid valve working principle

Table5-4: common fault and measurement of solenoid valve

Fault	reason	Measurement
No Direction changing	Coil broken	Replace the coil and check the electric wire connecting
	Piston seizure or blocked	Fix the piston
	Too big spring force or spring broken	Replace the spring
	Heavy friction of piston	Fix the piston
	Low pressure of control oil (if have pilot valve)	Heighten the control oil
	Jam in control oil line	Dredge the line
	Valve body distortion by tightly bolt driving	Re-fasten the bolt
	Valve piston blocked because of high oil temperature	Check the piping system, find out the reason then fix it
Coil broken	Bad insulated	Replace the coil
	Not coaxial of coil with valve piston	Reassembly
	Coil steel core fault	Fix or replace
	High voltage	Adjust the voltage
	Bad wire connecting	Re-connecting
Bad direction changing	Oil polluted	Clean the valve
	Big or small spring force	Replace the spring
	Polluted of coil	Clean the coil
	Big gap between piston and valve, or the control pressure too low	Adjust the gap and control pressure
Noise of coil action	Piston blocked or spring force too big	Fix or change the piston
	Bad touch of the coil Surface	Fix or change the coil
	Too big magnetic force	Change the suitable coil

5.2.3.3 RELIEVE VALVE

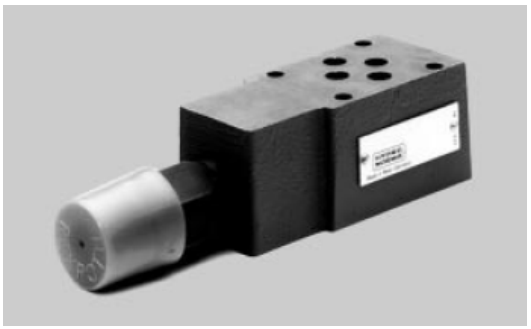


Fig.5-20：relieve valve

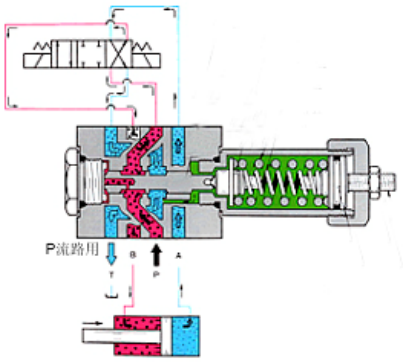


Fig.5-21：relieve valve work principle

Table 5-4：common fault and measurement of relieve valve

fault	reason	measurement
Pressure pulsation	Spring distortion or too soft	Change the spring
	Bad touch between poppet and valve seat or valve broken	Replace the poppet or remove the nut then push the bar for several times.
	Bad seal between steel ball and valve seat hole	Replace the steel ball ,fix the hole
	Poppet distortion	Replace or fix
Heavy leakage	Bad touch between poppet and valve seat	Replace the poppet
	Too big gap between poppet and valve seat	Replace the poppet or adjust the gap
	Loose tube joint or connector	Fasten the connector or bolt
	Oil polluted or jam in orifice	Clean the oil and dredge the orifice
Adjust fault	Spring broken or miss assembly	Check and replace
	Jam in orifice	Dredge the orifice
	Poppet blocked	Check and fix
	Mistake for the drain and suction port	Check and fix
	Leakage of the poppet	Check and fix
Heavy noise and vibration	Adjust bolt loosing	Fasten the bolt
	Spring distortion	Check and replace the spring
	Too tight assembly of poppet	Fix the poppet
	Poppet seizure	Replace the poppet
	Air in outlet port	Vent air
	Flux exceed permitted value	Replace high flux capacity valve
	Resonance with other valves	Slightly adjust the pressure value

5.2.3.4 PRESSURE REDUCE VALVE

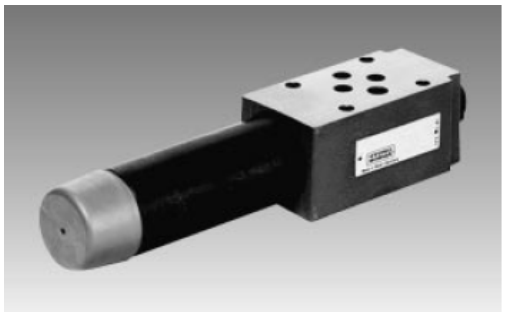


Fig.5-22： pressure reduce valve

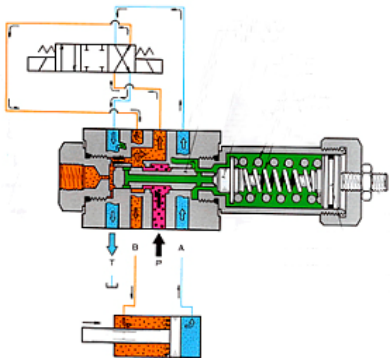


Fig.5-23： pressure reduce valve working principle

Table 5-6: common fault and measurement of reduce valve

Fault	reason	measurement
Pressure output pulsation	Air in the oil	Vent the air
	Jam in orifice	Dredge the orifice and clean the oil
	Spring distortion or blocked, piston move fault or bad assembly	Replace the spring
	Bad seal between steel ball and valve seat hole	Replace the steel ball or fix the poppet
	Piston move fault	Clean and fix
Low output pressure	Leakage on the top cover	Fasten the bolt or change the seal
	Bad seal between steel ball and valve seat hole	Replace the steel ball or poppet
Pressure fault	Oil plug in drain port have not been removed	Remove it and connect the hose
	Assembly fault of top cover, made the drain and suction line connected	reassembly
	Jam in orifice, poppet blocked or spring too soft	Clean the orifice, filter the oil and fix the spring or poppet

5.2.3.5 PRESSURE RELAY

Pressure relay is an important part in hydraulic system to connect electric and hydraulic. It is also a key part to control the system. With using time lapsing, the system pressure will change with the working order when equipment moves into steady period. So it is especially important to do the maintenance.



Fig.5-24： pressure relay

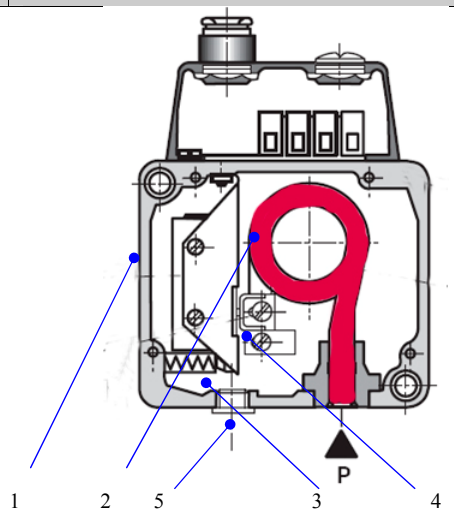


Fig.5-25： pressure relay working principle

1 cover 2spring 3 act control bar 4 switch

Figure 5-25 shows the spring-drive type pressure relay which uses spring as pressure transmit switch. The principle and manual as follows: pressure relay is consisted of cover1, spring 2 act control bar 3, two switch 4 and two adjust bolt 5. The switch will open or close with the pressure changing. The pressure is checked by spring2, when spring expand, 3 act control bar will push the switch4, so the electric line will be open or close. When pressure raise continuously the control bar will push the second switch then the relative electric line will open and close as well. There are 2 switches which are called high pressure point and low pressure point, when the switch acts, it will send relative signal to PLC. There are 6 connections port in electric box, grouped as 2 parts. In each part, one is public line (power line), the other two are signal line (normal open or normal close), choose the relative port to connect according to the working order.

The pressure relay's value could be adjusted by the position of the switch with control bar, so that through driving the adjust bolt to control the pressure relay act value.

Function applied in system

- control the motor

  In the disc brake or wheel brake system or other system, the pressure relay is used to control the motor to make the system pressure. When QC control on, if the system pressure is lower than the preset low value, then energize the motor. If the pressure is higher than the high preset value, the pressure relay will send the signal and stop the motor. So this procession keeps the pressure value maintaining at the working pressure of the brake. When energize the relative solenoid, relative brake will open or close to satisfy the working order.

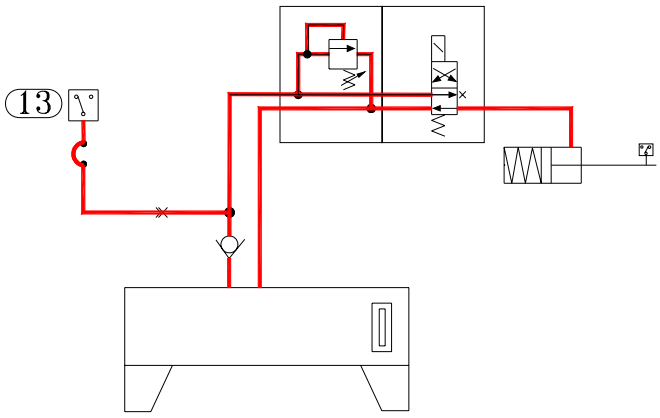


Fig.5-26：pressure relay used to control the motor

- used to send snag signal

As fig. 5-27 shows there are pressure relay (24A) mounted on every snag manifold. When snag happen, the pressure in snag cylinder will rise quickly. So the pressure relay will send snag signal when the snag pressure up to the preset value. So the QC will do an emergency stop.

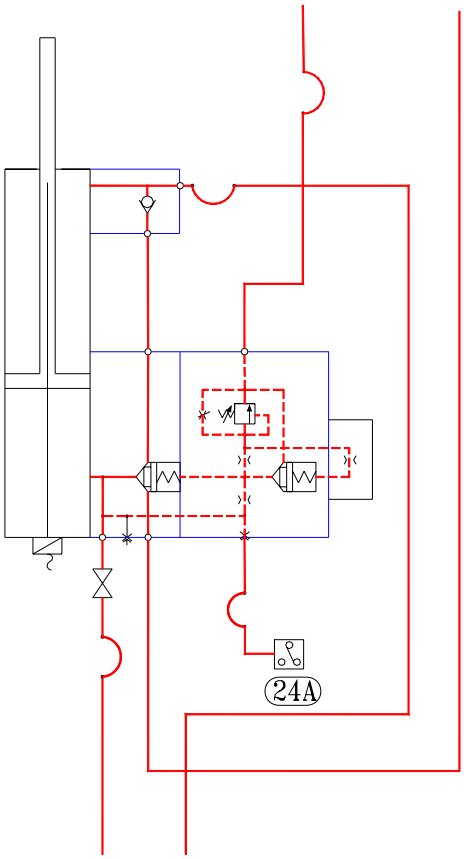


Fig.5-27：used to check the snag pressure

5.2.3.6 THROTTLE VALVE



Fig.5-28： throttle valve

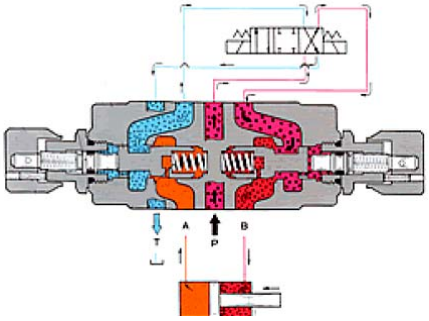


Fig.5-29： throttle valve working principle

- combine with check valve as a set:

  It is mounted on the wheel brake system outlet port. This set of valves makes the wheel brake opened quickly when high pressure pushed in, while clamping time will be limited by the throttle valve(as the damper)

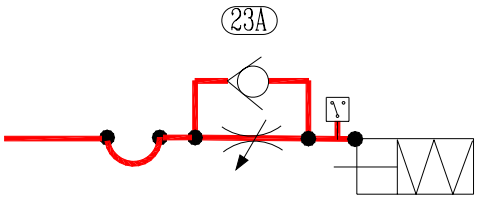


Fig.5-30： combined with the check valve

5.2.4 Other attention

5.2.4.1 REASONS FOR OIL TEMPERATURE RISEING AND ITS EFFECTION

General speaking, there is power loosing in system working, including pressure lost, volume lost and mechanical lost. All these energy are transfer to heat, which makes hydraulic system oil temperature rising. There are several reasons for this problem: high pressure, polluted oil, inner leakage in the hydraulic parts, low volume of the oil tank, cooler or heater working order fault. These will make a lot of damage effect on the system:

- The oil viscosity will decline and form heavy leakage, even will have an effect on the normal working.
- Make oil polluted, forms oxide substance. It will make jam in orifice or gap so that it affects the working.
- Make the gap between parts which have different expand coefficient be too narrow to be moved even be blocked.
- Make mechanical hot distortion, brake the normal accuracy. So it is important to control the system temperature strictly. Keep it between 30 to 60°C and do not exceed 70°C normally.

5.2.4.2 MEASUREMENT OF KEEPING SYSTEM TEMPERATURE:

- Reduce the pressure control valve's set value when it is too high to minimize the heat lost
- Enlarge the volume of the oil tank and install cooler when the volum is not sufficient and the condition is not satisfactory
- Choose proper hydraulic oil according to the environment when the accessory heat since oil viscosity is too high to increase the inner friction.
- When the piping tube's is reedy and bended, the flow friction will added and make the temperature rising. Replace the tube with bigger diameter and short size.
- Prevent the outside environment effect. Using anti-heat material to make isolation with outer side heat.
- Relieve the high pressure oil uncalled-for for a long time, modify the line or replace the unload measurement.

5.2.4.3 DIAGNOSIS OF HYDRAULIC SYSTEM FAULT .

There are various kinds of faults in hydraulic system The reason for faults maybe the product of multiple reason..It is forbidden to remove parts without reason. There are 2 kinds of fault diagnosis: direct experience judgment diagnosis and equipment testing diagnosis.

Direct experience judgment diagnosis:

1. 4-feelings diagnosis, service person use their touching, vision, hearing and smelling to judge the fault impossible reason through past experience.

- Touching: through the touching of parts or tube's vibration to judge the fault.
- Vision: there are phenomena like low power motion, unsteadily working, leakage or oil pollution can be watched by service person, then make some judgment through experience.
- Hearing: abnormal voice of working pump or motor, screaming of relieve valve or piping tube's vibration, all of these can be heard and then we can find out the fault reason.
- Smelling, judge whether the oil has been polluted and motor's heavy heat or not.

5.2.4.4 LOGIC ANALYSIS FOR WORKING OUT THE SYSTEM FAULT

Through fault phenomenon, using logic analysis, reduce suspicious objects, and then find out the fault parts. The basic steps are as following:

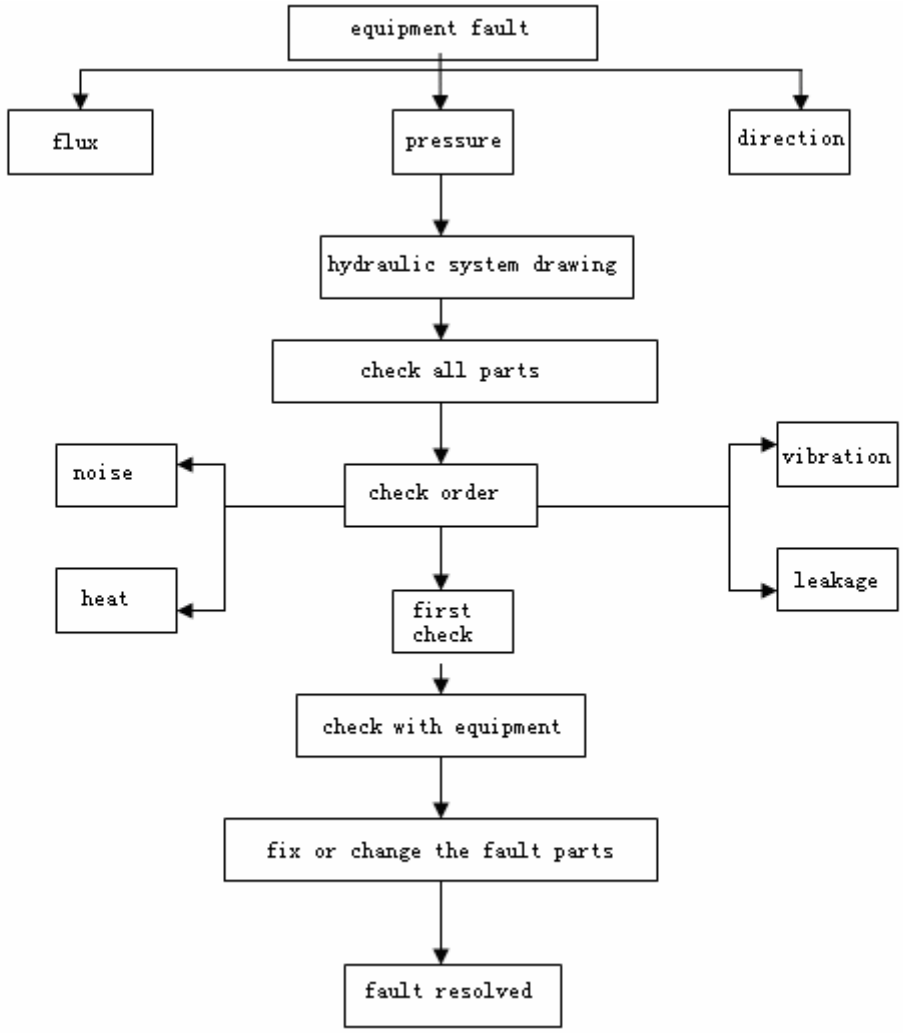


Fig.5-31： logic analysis of system fault

Hydraulic system's abnormal working can be led to 3 parts: pressure, flux and direction. Logic analysis is that according to the basic principle of hydraulic system, check the schematic drawing or relative parts to assure the function and effect, list the possible parts, and check them from easy to difficult.

Firstly check whether the parts choosing and assembly are appropriately or not; damage to outer signal or not; be attention to the foreboding such as: high temperature, noise, vibration or leakage, etc.

If there is no fault part found out by direct experience, then use test equipment to check again and again until find the fault part.

Then fix or replace the fault parts.

Think over the possible reason before restart the system, anticipated the potential danger and validate the judgment, then take relative measurement.

Equipment testing diagnosis:

Judge the fault through pressure flow, pollution and noise by equipment testing. There are kinds of test equipment:

Pressure testing equipment: tube type pressure gauge, accurate no-liquid pressure gauge, electric pressure gauge and baric pressure gauge, etc.

Flux testing equipment: volume flowmeter, throttle flowmeter, surface type flowmeter and turbine flowmeter, etc.

Pollution testing equipment:

Portable pollution tester, , laser pollution tester, ultrasonic pollution tester and electric pollution tester, etc.

Noise tester: sound-electric sensor and sound level tester.

More details of these equipments are shown in the instruction manual.

5.2.4.5 CLEANING OF HYDRAULIC SYSTEM

The same working hydraulic oil should be used instead of other liquid such as kerosene, gasoline, alcohol or steam to prevent rust of tube, hydraulic parts and seal broken.

In cleaning procession, the pump should be energized with oil heating at the same time. It is easy to remove the rubber dregs when temperature is about 50~80℃.

Using non-metal stick to beat the piping tubes continuously or not, it is better for remove the appendiculate substance.

Energizing motor intermittently is good for clean the pollution. Interval should be 10~ 30 mins.

Install filter in return line port, using thin filter net at first and thick ones at last.

Cleaning time of the system is 48~60 hours, the detailed timing is according to the environment complexity and pollution level.

tenance

Pump should still been energized after finishing until temperature become normal to prevent rusting led by outside humidity.

Drain out of the cleaning oil completely after cleaning.

5.3 GANTRY RAIL BRAKE SYSTEM

5.3.1 Schematic drawing and main parameter

This system is mounted on QC gantry drive system, each set on landside and seaside.

The system schematic drawing is shown below. Especially be attention to shut off valve (08B) and relieve valve (06); shut off valve is used to unload the pressure when changing oil or adjusting the pressure relay. So make sure of opening the shut off valve when doing the above processing and closing it after finishing.

Relieve valve is used as safety protection to restrict the system pressure always under the preset safe value. Check these valves during the daily main

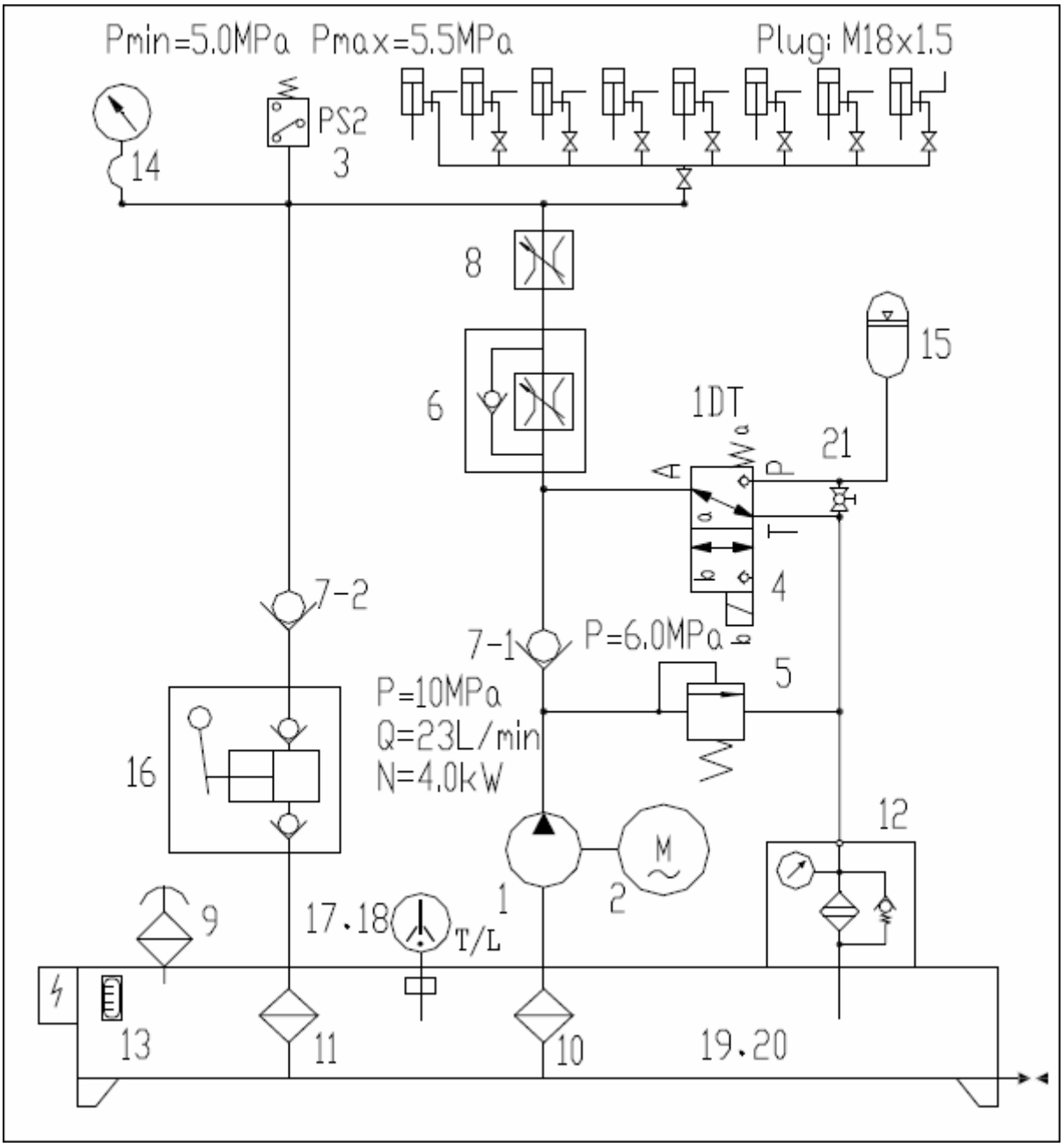


Fig.5-32：gantry rail brake system schematic drawing

Table 5-7: Parts list

序号	名　称	Accessory name
1	齿轮泵	Gear pump
2	电动机	Motor
3	压力传感器	Pressure sensor
	数显仪	Digital transmitter controller
4	电磁阀	Electromagnetic valve
5	溢流阀	Overflow valve
6	单向节流阀	One-way throttle valve
7-1	单向阀	Check valve
7-2	单向阀	Check valve
8	节流阀	Throttle valve
9	空气滤清器	Air filter
10	吸口过滤器(泵)	Meshy filter
11	吸口过滤器(手动泵)	Meshy filter
12	回油过滤器	High pressure filter
13	液位温度计	Liquid level gauge
14	压力表	Pressure gauge
15	蓄能器	Accumulator
16	手动泵	Hand pump
17	低位液位计	Gauge
18	超温温度计	Thermometer
19	油　箱	Tank
20	管路及附件	Pipeline and sccessories
21	板式二通阀	Plate two-way valve

5.3.2 instruction

- Wheel brake

Rail brake use spring to trig and hydraulic power to release, the detailed principle as following:

When pressure oil solenoid is pushed into brake cylinder controlled by solenoid valve, the oil will push the piston to compress the spring, and then the piston will drive the brake opened. When solenoid valve de-energized, hydraulic oil will be pushed back by the spring and brake will clamp automatically, build up the trig torque.

Control of the brake clamp time is an important problem, A check throttle valve is mounted on every outlet port of the wheel brake, which is used to control the clamp time. Every wheel brake cylinder has a limit switch, after all limit switches send out signal; the gantry is permitted to move.

When gantry stopping, the QC will have a great amount of inertia, it will still sliding for a distance. If the brake clamp is too early, the gantry will have a very big impact which is dangerous. So control the brake clamping time(normal) at1.5~5s s after de-energizing the gantry motor. But the range of adjustment is 0.3~10s

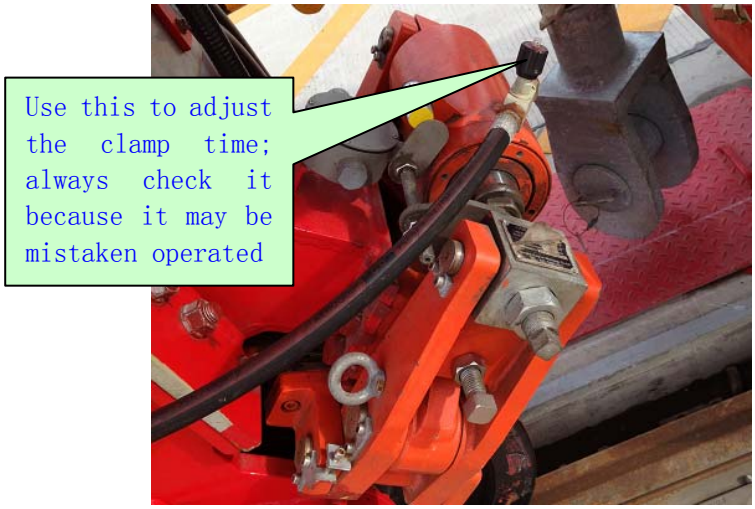


Fig.5-33：Wheel brake

- How to open rail brake manually

When there is some fault in the hydraulic system or no electric power, the brake can be released by hand:

- Close the shut off valve item 8。
- Pumping the hand pump to its working pressure
- Open shut off valve after finishing manual operation. Refresh the system at normal state.



Fig.5-34：wheel rail brake hydraulic unit

5.4 Machine house disc brake system

5.4.1 Schematic drawing and main parameter

There are 2 sets of disc brake systems, for boom and hoist system, using one hydraulic unit in machine house.

In this system, the hydraulic unit is used to control the boom or hoist emergency disc brake. For the relieve valve (06) (for the safety purpose), it is used to prevent the system overloading, solenoid valve (7A) are used to open or close the brake. Pressure relay is used to control the system pressure. Like rail brake system, shut off valve (08A) should be opened to unload the pressure when changing oil or adjusting pressure relay. But this valve should be closed after motion

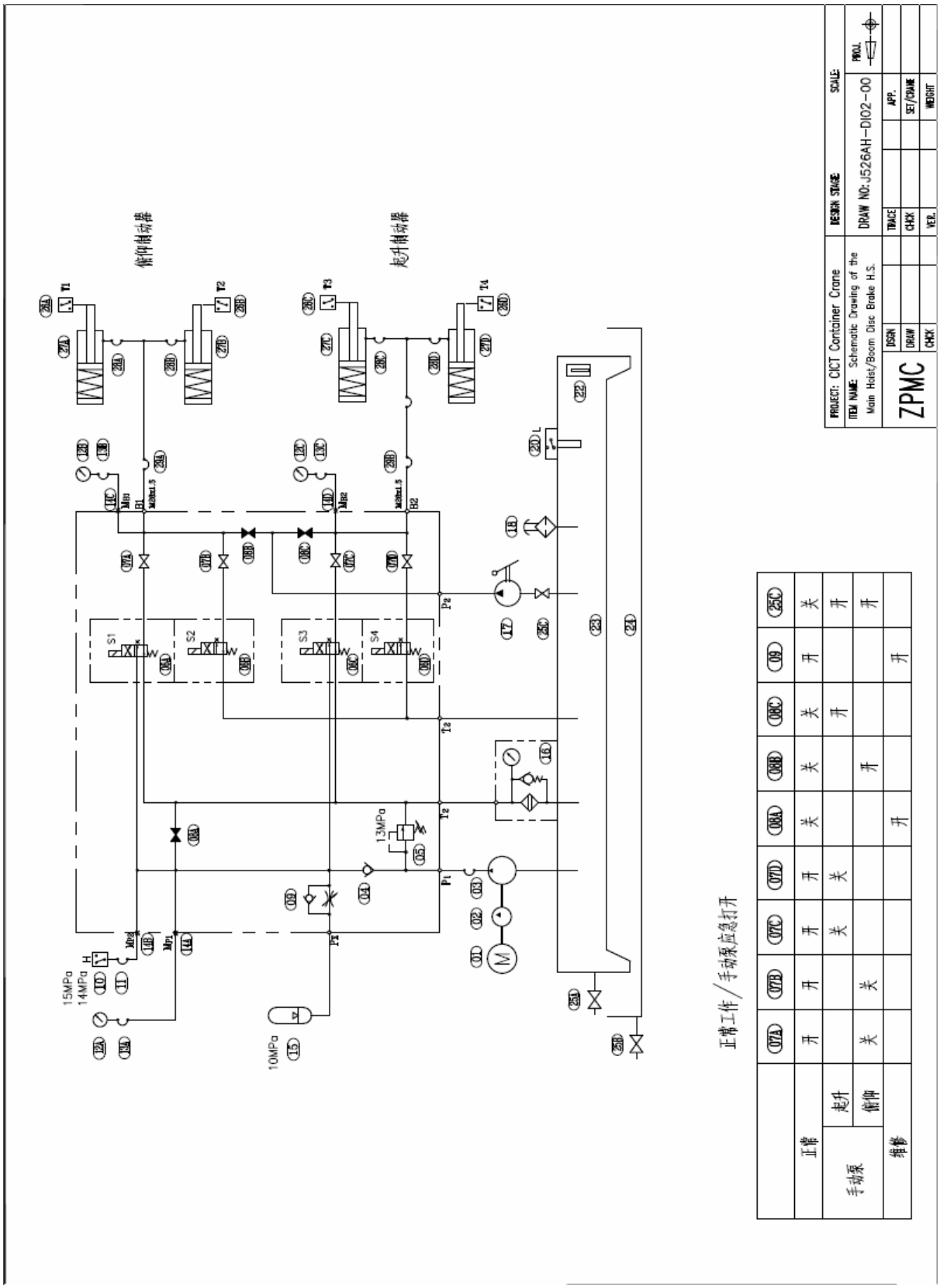


Fig.5-35: disc brake hydraulic system schematic drawing

Table 5-8: Parts list

List of Boom & Hoist Disc Brake Hydrauli System

NO.	SUN-NO. OR STDD	NAME	Q'TY	STYLE
1	HGG132S4A 5.5Kw1430rpm380v/50HzB5	Electrical Motor	1	COAL
2	ROTEX-28 ST98 1-19.05/1a-38	Coupling	1	KTR.SH
3	AZPF-1X-014RCB	Gear pump	1	REXROTH
4	CV1-10-P-0-15	Check valve	1	EATON
5	RV3-10-S-0-36	Relief valve	1	VICKERS
6	4WE10D3X/CW230N9K4	Solenoid valve	4	REXROTH
7	NV1-10-K	Shut off valve	4	VICKERS
8	NV1-10-R	Shut off valve	3	VICKERS
9	FCV7-10-C-0-FF/40	Check/throttle Valve	1	VICKERS
10	HED30A-30/200KL220	Pressure relay	1	REXROTH
11	6400-10.162-80.404-1000	Test hose	1	FUKA
12	213.53.063B25MpaG1/4A	Pressure gauge	3	WIKA.SH/TEC
13	6400-10.162-50.204-800	Test hose	3	FUKA
14	SKK20-M14x1.5-PB	Test connection	4	STAUFF
15	EHV10-330/90	Accumulator	1	OLARE
16	RFBN/HC110DC10ES1.0	Oil return filter	1	HYDAC
17	SB-1-20S	Hand pump	1	ZPMC
18	SPB-S-5-10-S080-A	Air filter	1	STAUFF
19				
20	SLWC-60-1-B250NC-01-01-1-T70NC-P0	Float switch	1	STAUFF
21				
22	SNA127B-S-T-12	Oil gauge	1	STAUFF
23	JXXXAH-	Tank	1	ZPMC
24	JXXXAH-	CATCH BASINS	1	ZPMC
25	G1/2"	Ball valve	3	AMK
26		Limit switch	4	ZPMC
27	SBB315-2200	Disc Brake	4	ZPMC
28	F381CACF-22-22-12-L	Hose	2	PARKER
29	F381CACF-15-15-8-L	Hose	4	PARKER

5.4.2 Instruction

- Instruction manual of disc brake

The disc brake use hydraulic power to release and spring force to trig. When the pressure oil is pushed into brake cylinder, oil will push the piston to compress the spring which makes the brake open.

There is an air venting process before using or adjusting the disc brake. Vent the air through the test coupling on the top of the brake cylinder.

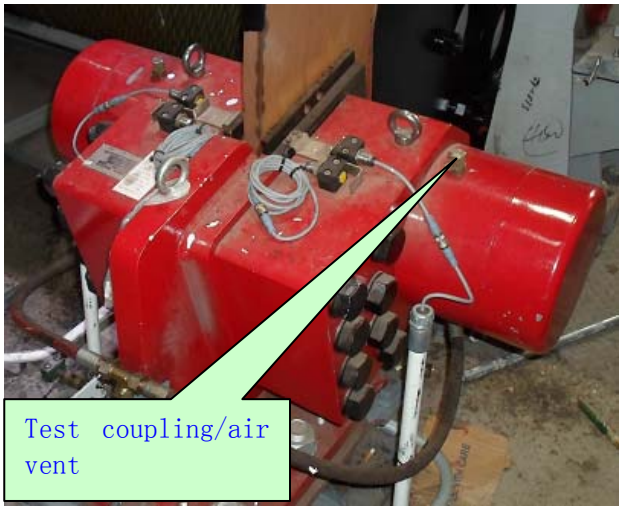


Fig.5-36： disc brake

- how to open cylinder manually

When there is some fault in the hydraulic system or no electric power, the brake can be releases by hand:

- Close the shut off valve 7C&7D(or 7A&7B), open the shut off valve 8C(or 8B) and 25C
- Pumping the hand pump to its working pressure, then the brake will open
- Open shut off valve (9B or 9A), close the shut off valve (8C), and refresh the system at normal state after finishing hand pumping.

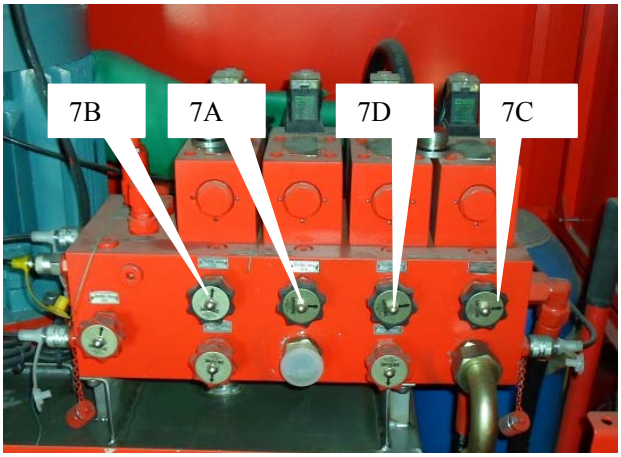


Fig.5-37： hydraulic unit

5.5 Multifunctional hydraulic system.

5.5.1 Schematic drawing and main parameter

The multi-function system unit is mounted on the back girder that is divided into 2 parts: T/L/S & snag protection and steel rope tension system.

The system main parameters are as followed:

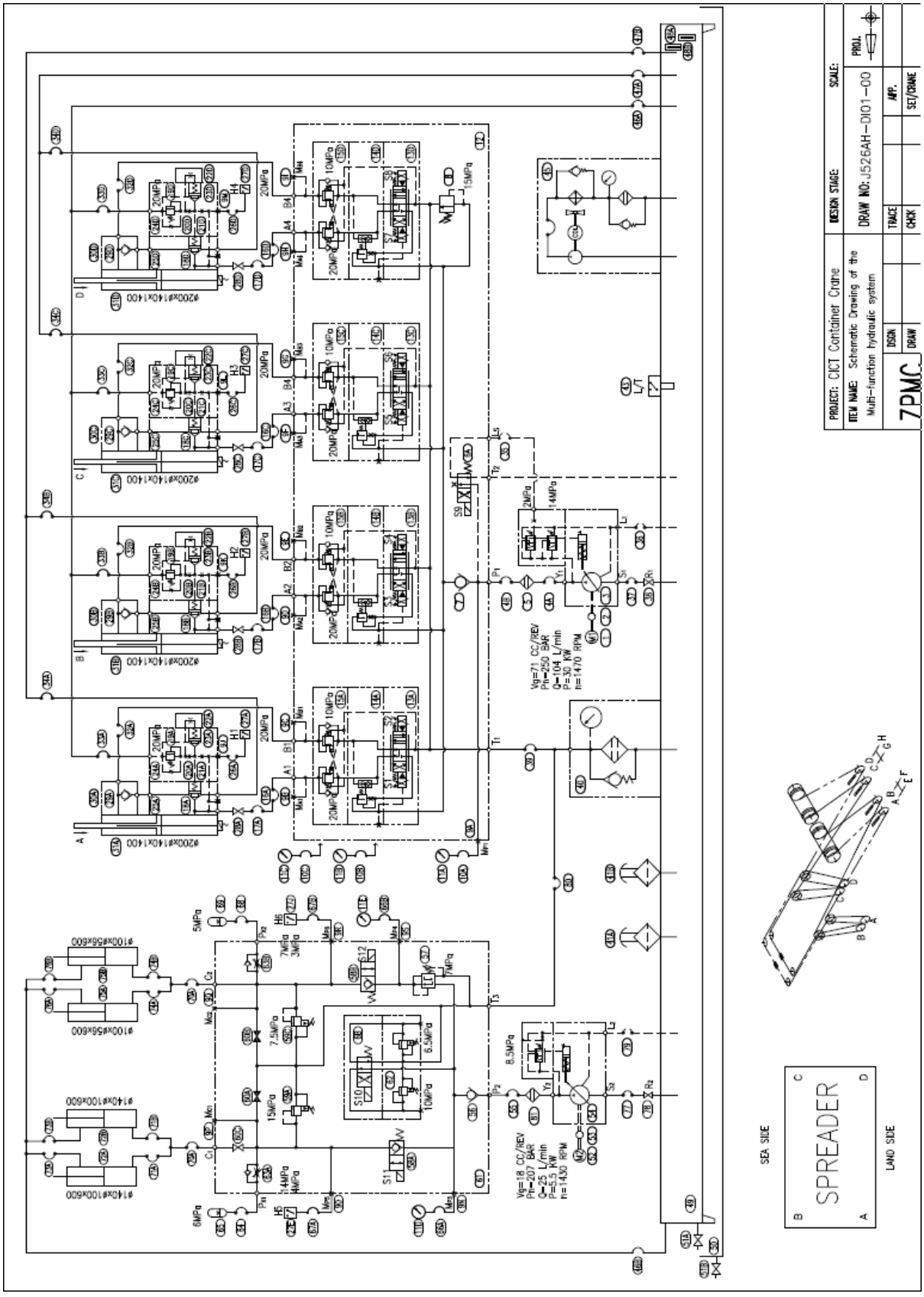


Fig.5-38: multi-function hydraulic schematic drawing

Table 5-9: Parts list

List of Multi-function hydraulic system

NO.	SUN-NO. OR STDD	NAME	Q'TY	STYLE
1	HGG200L4A 30Kw1480rpm400v/50HzB5	Electrical motor	1	COAL
3	A A10VSO71DRS/32R-VPB22U99-S2184	Piston pump	1	REXROTH
4	F3810639-16-16-16L	Hose	2	PARKER
5	DF BN/HC 240 TE 10 B 1.0/-B6	High pressure filter	1	HYDAC
6	4WE6D6X/EW230N9K4	Solenoid valve	2	REXROTH
7	CV2-20-P-0-30	Check valve	1	EATON
8	CG2V-6GW10	Relief valve	1	VICKERS
9	SKK20-M14x1.5-PB	Test connection	19	STAUFF
10	6400-10.162-50.204-1000	Test hose	3	STAUFF
11	213.53.063B25MpaG1/4A	Pressure gauge	5	WIKA.SH
12	JXXXAH-	Block I	1	ZPMC
13	4WRAE10W1-60-2X/G24N9KE31/F1V	Proportional valve	4	REXROTH
14	CWEA-LHN NBY/S	Counter balance valve	4	SUN
15	YFEJ-XHN-BA/S	Pressure Compensator	4	SUN
16	F3810639-8-8-8L	Hose	4	PARKER
17	F3810639-8-8-8L	Hose	4	PARKER
18	LC50DB20E7X/-104	Cartridge valve	4	REXROTH
19	DBDS6K1X/315	Relief valve	4	REXROTH
20	0.8/M6	Orifice	4	ZPMC
21	1.5/M6	Orifice	4	ZPMC
22	JXXXAH-	Block II	4	ZPMC
23	LC16DB20E7X/-004	Cartridge valve	4	REXROTH
24	DVP6-01.X	Shut off valve	4	HYDAC
25	JXXXAH-	Block III	4	ZPMC
26	6400-10.162-80.404-1100	Test hose	4	FUKA
27	HED30A-30/200KL220	Pressure relay	6	REXROTH
28	DVP10-01.X	Shut off valve	4	HYDAC
29	LC40B05E7X/	Relief valve	4	REXROTH
30	JXXXAH-	Block IV	4	ZPMC

31	Ø200xØ140x1400	Snag cylinder	4	HUAWU
32	F3810639-8-8-8L	Hose	4	PARKER
33	F3810639-8-8-8L	Hose	4	PARKER
34	F3811519-24-24-24L	Hose	4	PARKER
35	F3810639-6-6-6L	Hose	4	PARKER
36	F3810639-8-8-8L	Hose	1	PARKER
37	KXT-065-16NM	Flexible fitting	1	ZPMC
38	TECFLY-65-NBR-LS	Butter fly valve	1	ZPMC
39	F8811515-24-24-24L	Hose	1	JUNREN
40	RFBN/HC240DE10BS1.1	Oil return filter	1	HYDAC
41	ELFP7F10W1.0	Air filter	2	HYDAC
42				
43	SLWC-60-1-B500NC-01-01-1-T70NC-P0	Float switch	1	STAUFF
44				
45	SCA 2S/28/1.0/B/M/A/	Cooler	1	HYDAC
46	F3810639-8-8-8L	Hose	2	PARKER
47	F3811519-32-32-32L	Hose	2	JUNREN
48	SNA254B-S-T-12	Oil gauge	2	STAUFF
49	JXXXAH-	Tank	1	ZPMC
50	JXXXAH-	CATCH BASINS	1	ZPMC
51	G1/2″	Ball valve	3	AMK
52	HGG132S4A 5.5Kw1440rpm380v/50HzB5	Motor	1	COAL
53	ROTEX 28 ST98 1-22.2/1a-38	Coupling	1	KTR.SH
54	A10VSO18DR 31R PPA12N00	Piston pump	1	REXROTH
55	F3810639-10-10-8L	Hose	1	PARKER
56	D04B2-1.0N	Check valve	1	STERLING
57	DGMX2-3-PP-CW-3-40	Pressure reduce valve	1	VICKERS
58	SV13-10-CM-0-240AGH	Solenoid valve	2	VICKERS
59	RV3-10-S-O-36	Relief valve	2	VICKERS
60	NV1-10-R	Shut off valve	3	VICKERS
61	JXXXAH-	Block V	1	ZPMC
62	RPCC-LAN-ABX	Relief valve	1	SUN

63	FCV7-10-C-0-FF/40	Check/throttle valve	2	VICKERS
64	F3810639-8-8-8L	Hose	1	PARKER
65	EHV10-330/90	Accumulator	1	OLARE
66	6400-10.162-50.204-900	Test hose	2	FUKA
67	6400-10.162-80.404-1000	Test hose	2	FUKA
68	F3810639-8-8-8L	Hose	1	PARKER
69	EHV4-350/AC	Accumulator	1	OLARE
70	F3810639-8-8-8L	Hose	2	PARKER
71	F3810639-8-8-8L	Hose	2	PARKER
72	Φ140xΦ100X600	Rope tension cylinder	2	HUAWU
73	F3810639-8-8-8L	Hose	2	PARKER
74	F3810639-8-8-8L	Hose	2	PARKER
75	Φ100xΦ56X600	C tension cylinder	2	HUAWU
76	F3810639-8-8-8L	Hose	2	PARKER
77	KXT-040-16NM	Flexible fitting	1	PARKER
78	TECFLY-40-NBR-LS	Butter fly valve	1	ZPMC
79	F3810639-8-8-8L	Hose	1	PARKER
80	F3810639-8-8-8L	Hose	1	PARKER
81	DFBN/HC 110 TC10B1.0/-B6	Pressure oil filter	1	HYDAC

In this system, tension part and T/L/S snag part has their own respective loops.

In tension part, as shown in fig. 5-39, relieve valve is set at 15 Mpa and dual relieve valve is set at 10 and 6.5 Mpa. The main function of relieve valve (59) is providing pressure safety protection, reducing the impact on steel rope by catenary suddenly moving or stopping. Dual relieve valve(62) is controlled by solenoid valve, when boom up, S10 will energize , the system pressure will go down to 6.5 Mpa, after boom up , S10 will de-energized, system pressure will refresh to the pump set value.

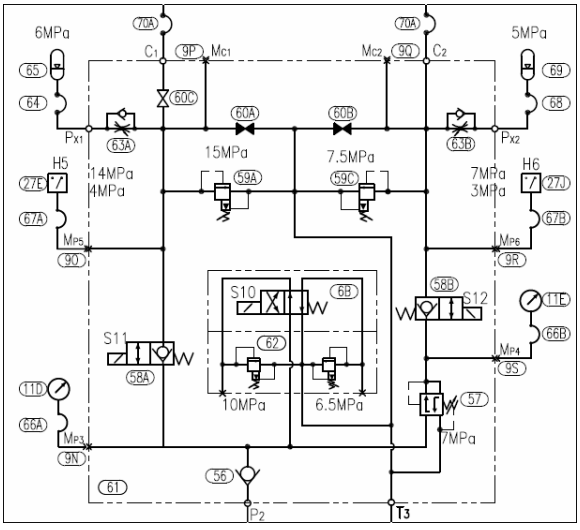


Fig.5-39： tension system



Fig.5-40： tension valve sets

Snag protection & T/L/S system consist of 2 parts : one is snag protection system which is mounted on snag cylinders. The other one is T/L/S spreader adjusts system which is mounted inside the unit.

- Snag protection system:

As fig. 5-41 shown, there are a check valve manifold and a snag protection manifold on the top side of the snag cylinder，the internal structure is shown as fig. 5-42, the snag system manifold includes a main plug-in logic valve, a pilot plug-in valve, 2 throttle valves, a relieve valve and 3 orifices. Shut off valve (24) is used unload the cylinder manually (drain the oil and retract the cylinder to "0" position. It should be closed at normal working state.

When snag happens, the pressure in non-rod chamber of the cylinder will rise quickly, when the value is up to the pressure setted, the pressure relay will send signal first, relieve valve (19) open, pilot valve will open as well which will drive the main logic plug-in valve open at the same time. At that time, both the two chamber of the cylinder are connected with each other to the tank. So the cylinder will retract quickly. This processing which is called mechanical sang protection, will protect the mechanical structure and steel rope. In this system, main plug-in valve is controlled by pilot valve and relieve valve. The shut off valve (16) should be opened normally or it will make the crane fail to do the T/L/S motion.

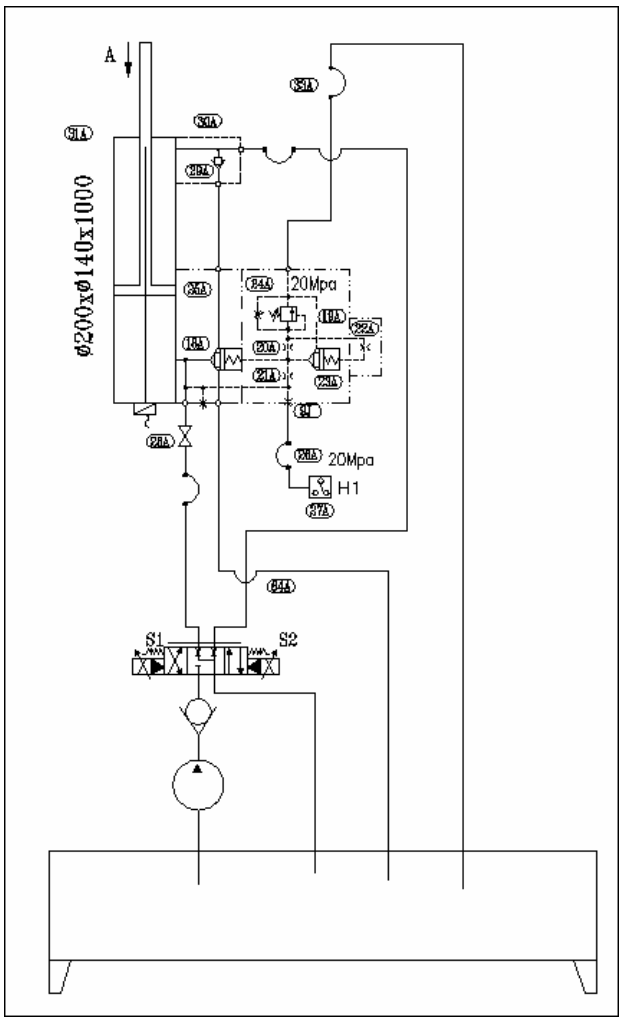


Fig.5-41： snag protection sys

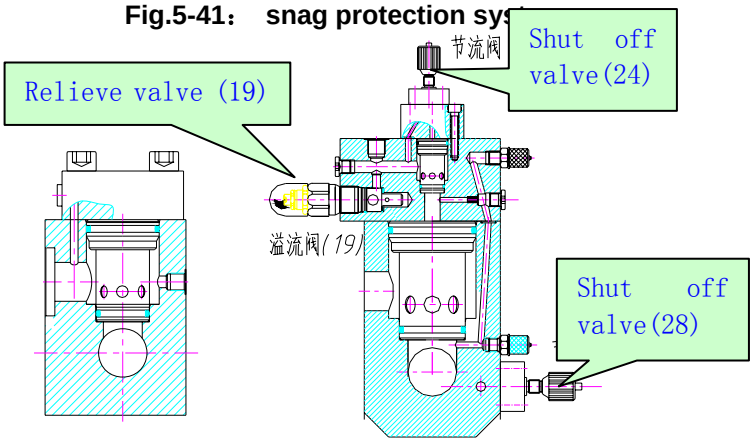


Fig.5-42：internal structure of snag manifold

- T/L/S system:

The main valves of this system include: proportional direction valve, pressure compensator valve and count-balance valve. The proportional valve control the cylinder's move direction and speed through flux and direction changing. Pressure compensator valve is used to keep a constant pressure difference between proportional valve's "P" port and "A" ,"B" port, which can make sure the proportional valve a steady working.

The count- balance valve can provide "back up" pressure to make the cylinder moving steady under loading. The solenoid valve will help the pump to unload the pressure when de-energized to do a standby power saving motion. While it is energized, the system pressure is controlled by pump setting.

There is a position sensor in each cylinder's piston, send electric signal to PLC to show the position of the barrel and piston, then the PLC will control the proportional valve through computer program automatically. So that makes a close loop control system of speed, direction and position.

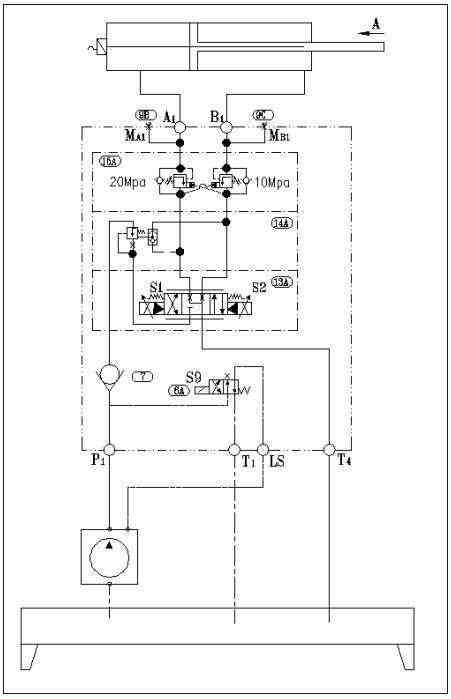


Fig.5-43： T/L/S system

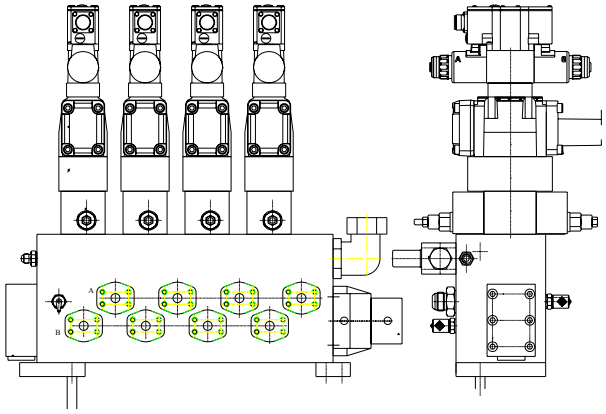


Fig.5-44：T/L/S system manifold

In **Fig.5-45：** the left one is T/L/S manifold and the right one is rope tension manifold. Make sure relative function of these manifolds is good to do the service and maintenance.

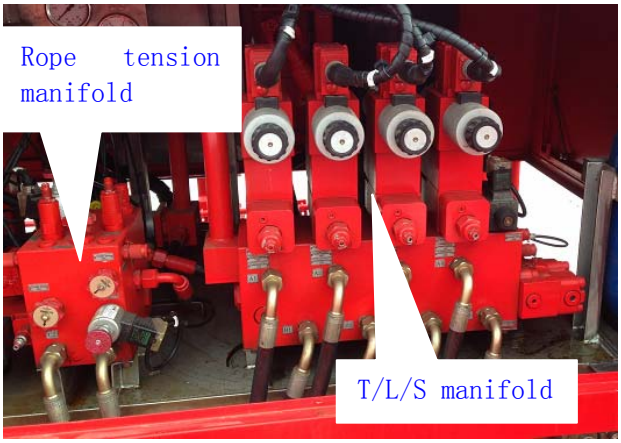


Fig.5-45：multi-function hydraulic unit

5.5.2 Instruction

- Snag cylinder and manifold working principle.

See fig. 5-46, at normal state, 4 piston rod extend to its normal position, the snag protection manifold mounted on cylinder's non-rod chamber outlet. The main valve (18) is plug-in logic valve, which controlled by pilot valve 23. meanwhile, pilot valve is controlled by safe relieve valve 19. the whole unload pressure is set by 19, when snag happen, the steel rope extend force will rise quickly, so anyone of the 4 snag cylinder's inside pressure go up to preset value, the pressure relay (27A-D) will send signal to PLC to stop the whole QC. At the same time, relieve valve will open the pilot valve. main valve will open as well. So the rod will retract, the piston will under the dynamic count-force to absorb the great deal of inertia through the snag stroke. This process guarantee the whole QC has a safe "soft landing" before emergency stopping. From the snag happening to QC stopping, the whole process is just in a very short time about 0.5 s, provide a highly safety to the crane.

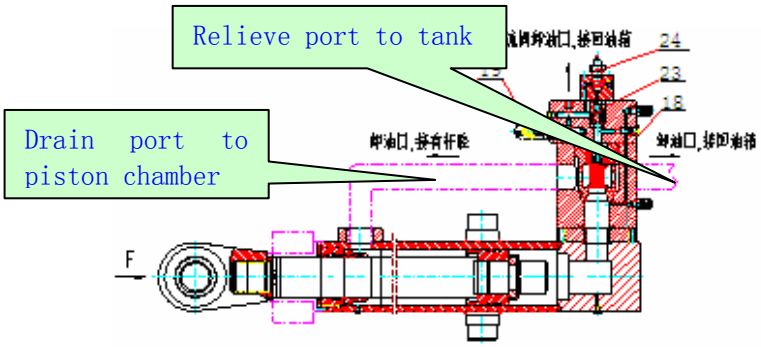


Fig.5-46：snag cylinder and manifold

- Test measurement of snag protection.

Firstly test the hydraulic part motion if snag

happens.

- When spreader do up motion, put off any one of the plug of the pressure relay. It is just simulate the real condition, to send a signal to PLC then the QC will emergency stop, and then put the plug back.
- Open **Fig.5-47：**, the snag cylinder will unload and retract. When the limit switch is out of the home position. Close the shut off valves(24), the cylinder will stop, this simulate the cylinder retract procession.
- Do down motion of the spreader for a safety distance. The spreader can't do up motion.
- Then the driver push the "control on reset" button, the cylinder will go back automatically. The QC will be at normal state again.

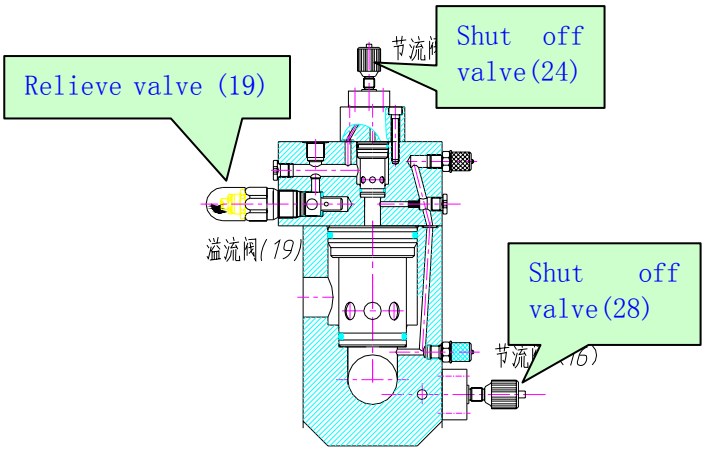


Fig.5-47：snag manifold

- Fault diagnosis and measurement of snag manifold.

  Most of the phenomenon of manifold fault is sliding

  - Close 1# valve shown in fig.5-49(this valve should be normal open), if cylinder stops sliding, it means the fault is in unit(not in manifold), then check the relative valves in units.
  - If 1# valve is closed and cylinder is still sliding, and then check whether 2# valve is close or not, it should be normal close.
  - If 2 # valve is closed, the cylinder is still sliding, then retract the cylinder to the end(open the 2#), exchange the 2# valve, push the control on reset button, the cylinder should extend to normal home position. Check the cylinder sliding or not.
  - Generally speaking, the manifold fault is about the pollution problem. The slight pollution can be solved by followed steps:
  - Make sure 1# valve is normal open, and then open the 2# valve slowly, so the cylinder will retract slowly, the speed is controlled by the 2# valve. Under the safety limit, make the speed faster, when cylinder retract for 400 mm, close 2# valve, then push the "control on" button, the retracted cylinder will extend to its normal position. Then repeat the above steps. This procession is used to wash out the polluted impurity by high pressure flow.
  - After the above steps reassembly the 3# hose joint if joint still slide. Check whether there is oil flow out from the manifold, which means the fault is about the pilot valve manifold. Then follow steps from 6~10, if not so, follow steps 11,12:
  - 6. Power off the QC, retract the 5# valve in fig. 5-51, remove the 5# manifold, check and clean it, then arrange it back. Or change the whole manifold with another one.
  - 7.remove 4# manifold , clean it (there is a cover should be remove, then it can be cleaned completely)
  - 8. Drive loose the inner hexagon bolt（6# in fig.5-49）, the manifold 7# can be removed . Clean the plug-in valve (8# valve in fig. 5-48) and orifice (9# in fig.5-48), arrange then back after finishing cleaning. Pay attention to the o ring between 7# manifold and 10# manifold in fig. 5-51.
  - 9. Arrange back 4# manifold. Pay attention to the o ring between 7# manifold and 4# manifold in fig. 5-51
  - 10. Power on the QC, cylinder will extend to home position automatically, then power off the QC, check the cylinder still sliding or not.

- 11. Drive loose the inner hexagon bolt （6# in fig.5-49）, the manifold 7# can be removed . Clean the plug-in valve (11# valve in fig. 5-48) in 10 # manifold. Then arrange back after cleaning. Pay attention to the o ring between 7# manifold and 10# manifold in fig. 5-51.
- 12. Power on the QC, cylinder will extend to home position automatically, then power off the QC, check the cylinder still sliding or not.

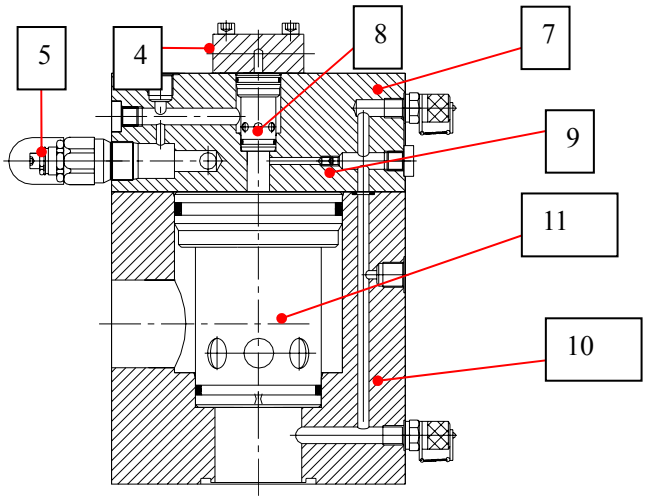


Fig.5-48：snag manifold

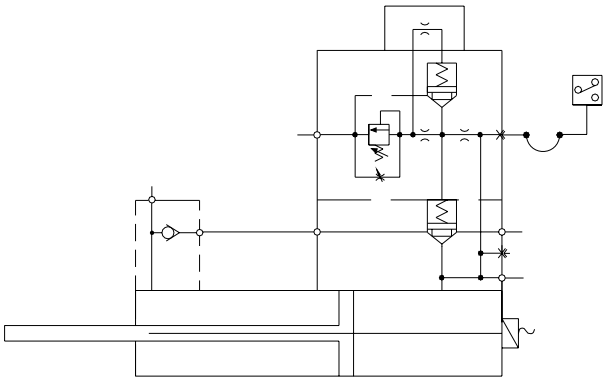


Fig.5-50： schematic drawing of snag protection principle

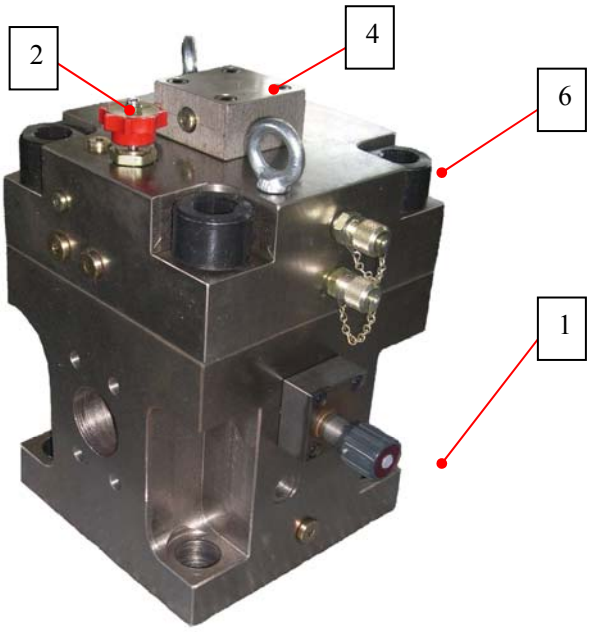


Fig.5-49：snag manifold

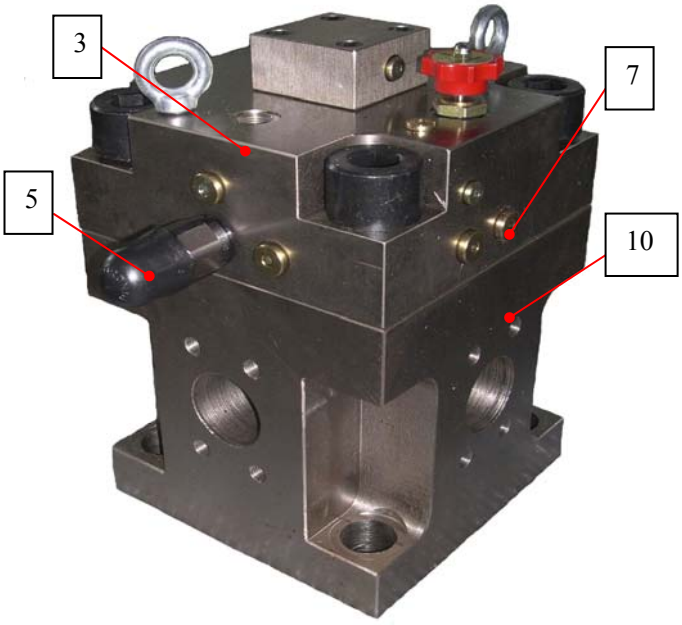


Fig.5-51：snag manifold

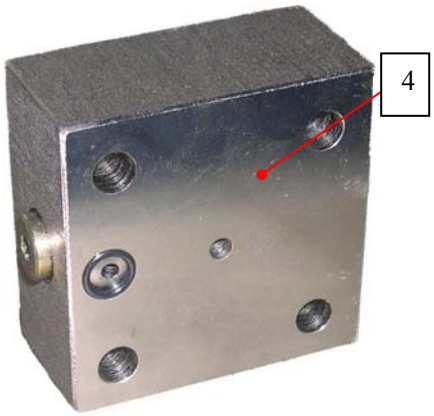


Fig.5-52：pilot valve manifold cover

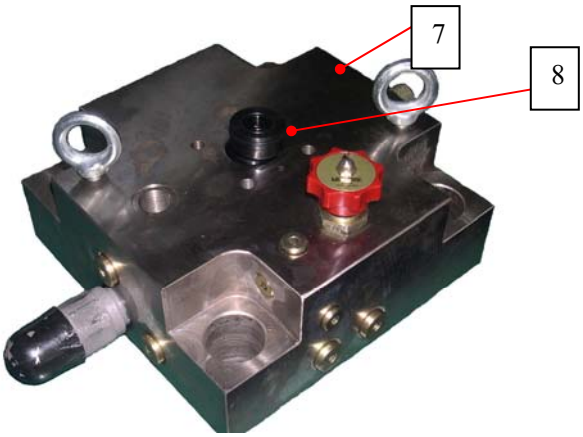


Fig.5-53：pilot valve manifold

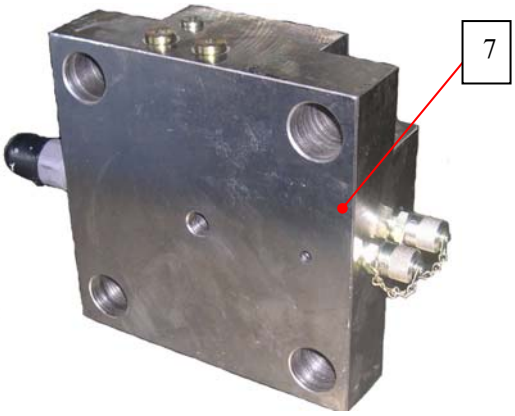


Fig.5-54：pilot valve manifold

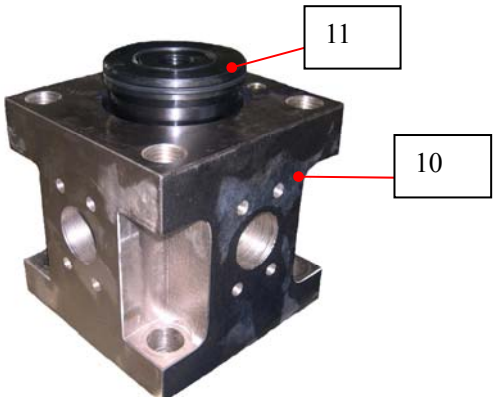


Fig.5-55：main snag protection manifold